

This is Part I of the overall course project which is based on ASSIMULO/SUNDIALS in your own Python working environment. In this part, we consider unconstrained ordinary differential equations and investigate the notion of order and stability by considering a spring pendulum example.

Please describe your observation in a report form. It should include

- a description of the problem at hand,
- a description of your tests, i.e., what is tested and why...,
- a description of your computational settings in such a way so that the numerical results are reproducible,
- your own interpretation of the results.

Make sure to describe your experiments and results thoroughly, and justify your statements carefully. Consider the tasks listed below as suggestions for experiments. Feel free to design others as long as they lead to a concluding statement. This assignment has 4 tasks.

Task 1

We consider the elastic pendulum given by the following ODEs

$$\dot{y}_1 = y_3 \quad (1)$$

$$\dot{y}_2 = y_4 \quad (2)$$

$$\dot{y}_3 = -y_1 \lambda(y_1, y_2) \quad (3)$$

$$\dot{y}_4 = -y_2 \lambda(y_1, y_2) - 1 \quad (4)$$

with $\lambda(y_1, y_2) = k \frac{\sqrt{y_1^2 + y_2^2} - 1}{\sqrt{y_1^2 + y_2^2}}$. It corresponds to a mathematical pendulum where the connecting bar is replaced by a linear spring with elastic constant k (Hooke's law). The mass, nominal (undeformed) pendulum length, and the gravitational constant are taken to be 1.

Set up the problem (especially the RHS-function) in Python. To verify the problem is set up properly, use a relatively small k in this task. First, use Assimulo's `Explicit_Problem` class to define this model problem; then use CVODE to solve the problem. Plot the results.

Task 2

Implement in Assimulo BDF-4 (*). Use Newton's iteration to solve the nonlinear system of equations in every time step. You can use the solution at the previous time step as the initial point (i.e., predictor) in the iterations for the current time step. The provided code (`BDF2_Assimulo.py`) for BDF-2 will give you an idea on how to make such an Assimulo implementation.

Task 3

A large value of k refers to high natural frequencies in this model. These frequencies are excited, when the initial condition corresponds to a (slightly) stretched spring, i.e., $\sqrt{y_1(0)^2 + y_2(0)^2} > 1$. In this task, we study the influence of this parameter on the choice of time integration methods.

Test three methods in Assimulo for various values of k (including large ones): the BDF-2 solver with fixed-point iteration (`BDF2_Assimulo.py`), the BDF-4 solver with Newton's iteration (your implementation), and the explicit Euler method. What do you observe in terms of accuracy and stability? Can you practically verify the messages from stability diagrams?

Task 4

Repeat the experiments in Task 3 with CVODE. Test also the influence of ATOL, RTOL, MAXORD, and choices for discretization on the computational performance, for both the lowly and highly oscillating scenarios.

*See also https://en.wikipedia.org/wiki/Backward_differentiation_formula